

Report For Fingerprint Quality Assessment

1. What I used for blur metric is the Variance of Laplacian with threshold value as 10.0.

Initially I selected it by seeing the common open CV examples, then I observed sharp images consistently produced variance values higher than 10 and vice versa for blurred images.

2. Ridge Quality metric was hard to achieve. I implemented it using the single Gabor filter but a very dark image also produces a high ridge score which is not satisfiable.

- It is measuring the variance over the entire image along with the background.

3. It is an algorithm for assessing the fingerprint quality. It is primarily used for contact based sensors, where the position, lighting, scale etc are fixed. but when we come to phone cameras the story is different

varied lighting, varied position; background etc also matters.

4. • Finger orientation : is it rotated or tilted.

- Camera distance : which contribute to the quality space of the finger.

- Wet or Dirty finger detection : as dirt and mud hinders the ridge pattern.

5. • Capture multiple images and select highest quality.

- Maybe applying stronger image enhancement techniques